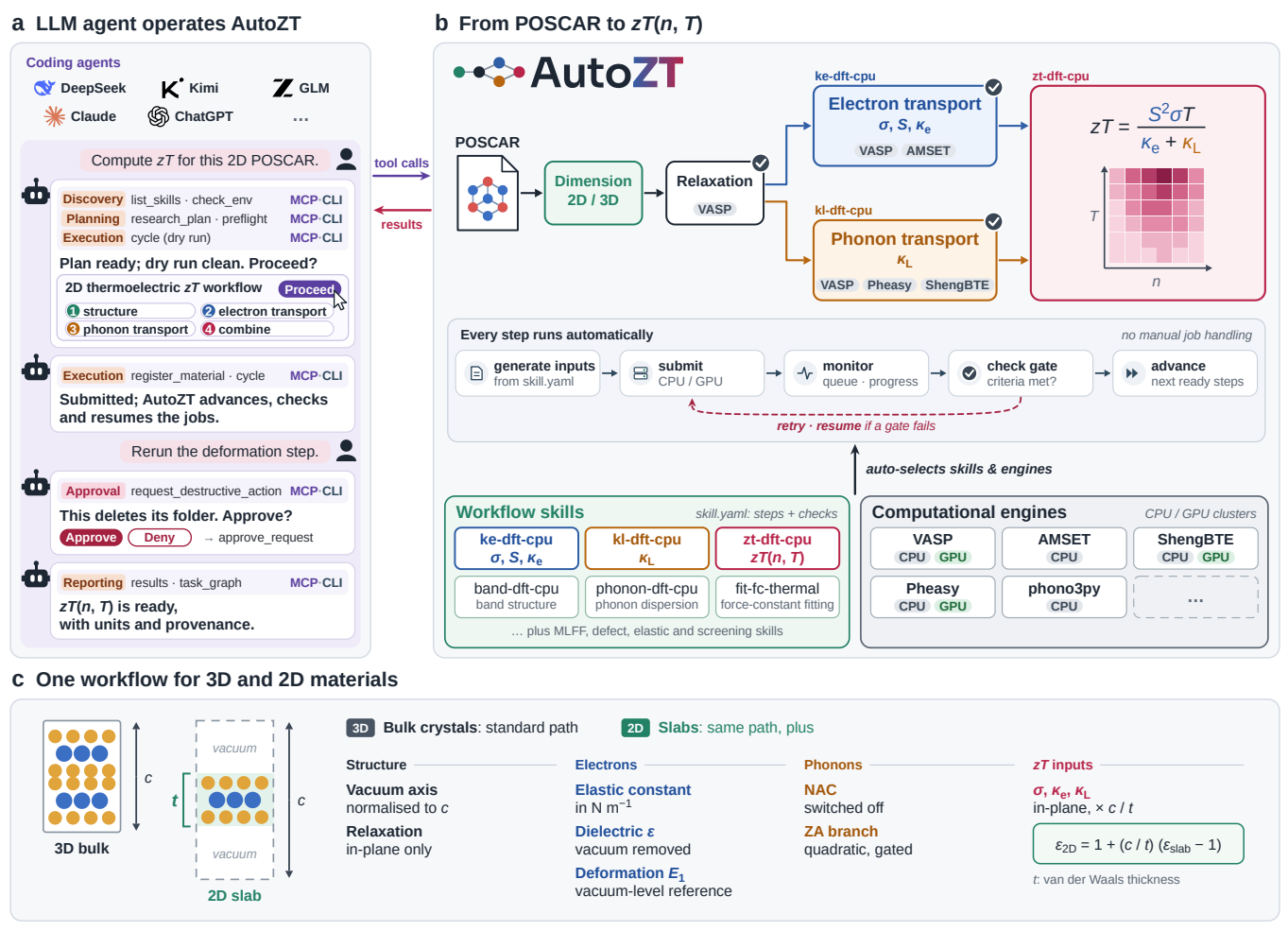

a LLM agent operates AutoZT
Coding agents
DeepSeek
Kimi
GLM
Claude
ChatGPT
...
Compute zT for this 2D POSCAR.
tool calls
results
Discovery list_skills · check_env MCP·CLI
Planning research_plan · preflight MCP·CLI
Execution cycle (dry run) MCP·CLI
Plan ready; dry run clean. Proceed?
2D thermoelectric zT workflow
Proceed
1 structure
2 electron transport
3 phonon transport
4 combine
Execution register_material · cycle MCP·CLI
Submitted; AutoZT advances, checks and resumes the jobs.
Rerun the deformation step.
Approval request_destructive_action MCP·CLI
This deletes its folder. Approve?
Approve
Deny
→ approve_request
Reporting results · task_graph MCP·CLI
zT(n, T) is ready, with units and provenance.
b From POSCAR to zT(n, T)
AutoZT
POSCAR
Dimension 2D / 3D
Relaxation
VASP
ke-dft-cpu
Electron transport
σ, S, κe
VASP AMSET
kl-dft-cpu
Phonon transport
κL
VASP Pheasy ShengBTE
zt-dft-cpu
zT = S²σT / (κe + κL)
T
n
Every step runs automatically
no manual job handling
generate inputs from skill.yaml
submit CPU / GPU
monitor queue · progress
check gate criteria met?
advance next ready steps
retry · resume if a gate fails
auto-selects skills & engines
Workflow skills
skill.yaml: steps + checks
ke-dft-cpu σ, S, κe
kl-dft-cpu κL
zt-dft-cpu zT(n, T)
band-dft-cpu band structure
phonon-dft-cpu phonon dispersion
fit-fc-thermal force-constant fitting
... plus MLFF, defect, elastic and screening skills
Computational engines
CPU / GPU clusters
VASP CPU GPU
AMSET CPU
ShengBTE CPU GPU
Pheasy CPU GPU
phono3py CPU
...
c One workflow for 3D and 2D materials
c
t
vacuum
vacuum
c
3D bulk
2D slab
3D Bulk crystals: standard path
2D Slabs: same path, plus
Structure
Vacuum axis
normalised to c
Relaxation
in-plane only
Electrons
Elastic constant
in N m⁻¹
Dielectric ε
vacuum removed
Deformation E1
vacuum-level reference
Phonons
NAC
switched off
ZA branch
quadratic, gated
zT inputs
σ, κe, κL
in-plane, × c / t
ε2D = 1 + (c / t) (εslab − 1)
t: van der Waals thickness